

BEA: Offering personalized, swift and secure banking services through data analytics

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ABOUT BEA

By building a unified data lakehouse and analyzing data in Google Cloud, BEA is able to realize its vision of data-driven decision making, enhancing its customer experience while ensuring full regulatory compliance.



About BEA

The Bank of East Asia, Limited ("BEA") is a leading Hong-Kong-based financial services group, the Bank is on its digital transformation journey. Founded in 1918, BEA provides a comprehensive range of financial services to customers through its extensive network covering Hong Kong, Chinese Mainland, Macau, Taiwan, Southeast Asia, the US and the UK. Offering wide-ranging services to individual and corporate customers, BEA is enhancing its customer experiences through the adoption of innovative technologies such as the cloud.

Google Cloud results

- Ensures data consistency for improved cross-departmental collaboration with Cloud Composer
- Supports meticulous data access control with policy tags in BigQuery
- Enables centralized security monitoring and threat detection with Cloud Monitoring and Cloud Logging

Data |
time
signif
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Industries: Financial Services & Insurance

Location: Hong Kong

(<https://cloud.google.com/composer/>),
Logging (<https://cloud.google.com/logging/>),
Google Cloud (<https://cloud.google.com/>)

BigQuery (<https://cloud.google.com/bigquery/>) and
IAM (<https://cloud.google.com/iam/>), Cloud

Cloud Composer
(<https://cloud.google.com/composer/>)
Center/)

Logging (<https://cloud.google.com/logging/>)

Monitoring (<https://cloud.google.com/monitoring/>)

Cloud IAM (<https://cloud.google.com/iam/>) help.

Cloud Security Command Center
(<https://cloud.google.com/security-command-center/>)

Digital transformation is changing the banking industry in new and revolutionary ways. Besides providing faster and more convenient services, banks are also using new technologies such as big

data and blockchain to transform their financial products and reinforce transaction security.

With a vision of being the trusted and preferred banking partner in Greater China and beyond, The Bank of East Asia, Limited

(<https://www.hkbea.com/html/en/index.html>) ("BEA") is on its digital transformation journey. Founded in 1918, BEA is a leading banking and financial services provider in Hong Kong, with branches across Chinese Mainland, Taiwan, Singapore, the US and the UK. Offering wide-ranging services to individual and corporate customers, BEA is enhancing its customer experiences through the adoption of innovative technologies such as the cloud.

"We're dedicated to ensuring every customer experience at BEA is positive by delivering best-in-class financial products and services. For us, digital transformation is an important means to achieve this goal," explains Stephen Leung, GCIO, GM and Head of Technology and Productivity Division at BEA.

As part of its digital transformation strategy, BEA wanted to implement a data-driven culture, with a fully digitized data platform as the cornerstone of this plan. Its team has mapped out a three-phase project to migrate its legacy data processing system into a centralized data platform named Metabase that supports reporting, analytics and prediction. Launched in 2022, the bank deployed the platform in Google Cloud (<https://cloud.google.com/>), due to the cloud's high-performance data processing tools. It

also allows BEA to be compliant with the strict security standards of the banking industry.

"As a financial service provider dealing with a large amount of data every day, data processing capability and security are equally important to us," notes Leung. "Google Cloud not only has exceptional data analytics offerings, but also has great emphasis on security through diverse approaches and tools."

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Enhancing customer experiences and operational efficiency with a unified data lakehouse

In 2023, BEA completed the first phase of its Metabase project by establishing a data lakehouse in BigQuery (<https://cloud.google.com/bigquery>). The team uses BigQuery to process structured core banking data. Throughout the rollout process, Google Cloud collaborated closely with BEA for a smooth transition.

"When we rolled out the first phase of our project, the Google Cloud team provided best practice guidelines, easy-to-use SQL conversion tool and expert advice, which made it easier for our team to keep up with the schedule and achieve our migration goals," says Leung.

To ensure data consistency across the organization, BEA also leverages Cloud Composer (<https://cloud.google.com/composer>) to orchestrate data. Before adopting Google Cloud, the various departments in BEA did not adhere to the same standards for data structure and format, which led to additional verification efforts and potential errors when data was shared across departments. With Cloud Composer, BEA can improve cross-departmental collaboration.

Through BigQuery, BEA has increased data visibility and cultivated a data-driven culture. By integrating business intelligence tools with BigQuery, its employees can easily retrieve data analytics reports that assist in their decision-making process. And as

BigQuery reduces data processing time significantly, BEA can conduct near real-time analytics, improve operational efficiency and enhance customer experiences by quickly responding to customer needs based on near real-time insights.

"BigQuery has not only helped us swiftly consolidate all our data in one place, but also facilitates the implementation of our data-driven strategies with its powerful analytics capabilities," notes Leung. "As a result, we've optimized our operations, customer services, as well as costs due to less time spent on data management."

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Safeguarding data security from access control to threat detection

When it comes to the tools for building Metabase, stringent security is one of the bank's primary requirements. According to Leung, Google Cloud has helped the bank fulfill the regulatory compliance requirements by the Hong Kong Monetary Authority and other government agencies with ease. Besides offering encryption with customer-managed encryption key for data at rest and in transit on its cloud platform, Google Cloud also possesses the required certifications for local and global compliance, such as ISO/IEC 27001 and PCI DSS.

Furthermore, Compliance Reports Manager (<https://cloud.google.com/security/compliance/compliance-reports-manager>)

provides BEA with seamless access to the compliance resources in Google Cloud.

To further minimize security risks, BEA taps on Cloud Audit Logs (<https://cloud.google.com/audit-logs>) to track user activity, Security Command Center (<https://cloud.google.com/security-command-center/>) to identify security misconfigurations, and Cloud Logging (<https://cloud.google.com/logging/>) and Cloud Monitoring (<https://cloud.google.com/monitoring/>), integrated with its monitoring and analysis system to deploy a central platform for real-time security monitoring and threat detection. The BEA team can therefore react timely to any security issues.

As for data integrity, BEA leverages Identity and Access Management (<https://cloud.google.com/iam>) and policy tags

(<https://cloud.google.com/bigquery/docs/best-practices-policy-tags>)

in BigQuery to implement meticulous access control. This way, the bank is able to establish efficient data governance and effortlessly maintain data consistency across the same dashboards or reports.

"Google Cloud provides diverse tools to safeguard the security of our data platform," adds Leung.

"Besides benefitting from the highly secure infrastructure of Google Cloud in compliance with global regulations, we've also adopted different advanced security features of Google Cloud to further strengthen data security and integrity."



Supporting smarter operations and services with AI

Currently in the second phase of its Metabase project, BEA is incorporating more internal and external data into its data platform for smart analytics. The bank plans to use Vertex AI (<https://cloud.google.com/vertex-ai/>) and AutoML

(<https://cloud.google.com/automl/>) to build machine learning models that can segment its customer base, perform fraud detection and anticipate trends. Through predictive analytics, BEA wants to provide more personalized services and optimize its marketing strategies. It will also leverage the generative artificial intelligence (AI) technology of Google to create new data for scenario testing and conduct more effective marketing campaigns.

Leung says, "With its excellent data analytics capabilities and comprehensive security tools, Google Cloud has helped us move an important step forward in our digital transformation journey by building a centralized data platform. In the next phase, we'll unleash the full potential of our data platform to continue upgrading our banking services."

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